KOSZUL-REGULAR, H_1-REGULAR AND QUASI-REGULAR IMMERSIONS OF SPACES ARE STABLE UNDER FLAT BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10583-10593 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 10595-10602, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Definitional-transfer template; the retained slogan environment is part of the pinned statement and shipped as such.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's definition section for the immersion flavours and flat morphisms of spaces; Divisors, lemma flat-base-change-regular-immersion (the schemes/ring-side stability of regular sequences under flat base change).

Source reference. Original lemma 06BN: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10530–10546. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 067P: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10602–10602. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03MK: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10602–10602. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Regular immersions are stable under flat base change. Let S be a scheme. Let $i : Z \to X$ be a Koszul-regular, H_1-regular, or quasi-regular immersion of algebraic spaces over S. Let $X' \to X$ be a flat morphism of algebraic spaces over S. Then the base change $i' : Z \times_X X' \to X'$ is a Koszul-regular, H_1-regular, or quasi-regular immersion.*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Via Definition 06BN (and the definition of a flat morphism of algebraic spaces in Morphisms of Spaces, Section 03MK) this lemma reduces to the case of schemes, see Divisors, Lemma 067P. □